Electronic Circuit Design

Lesson 2: Operational Amplifier Fundamentals II

Key Terms

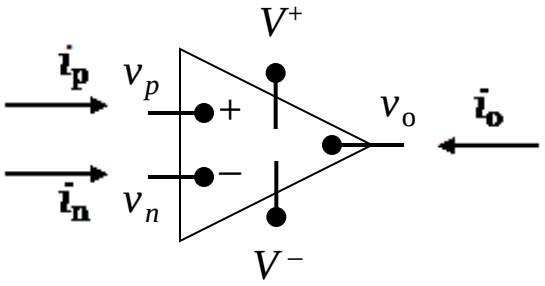
- negative feedback
- gain desensitivity
- distortion reduction
- loop gain
- op amp powering

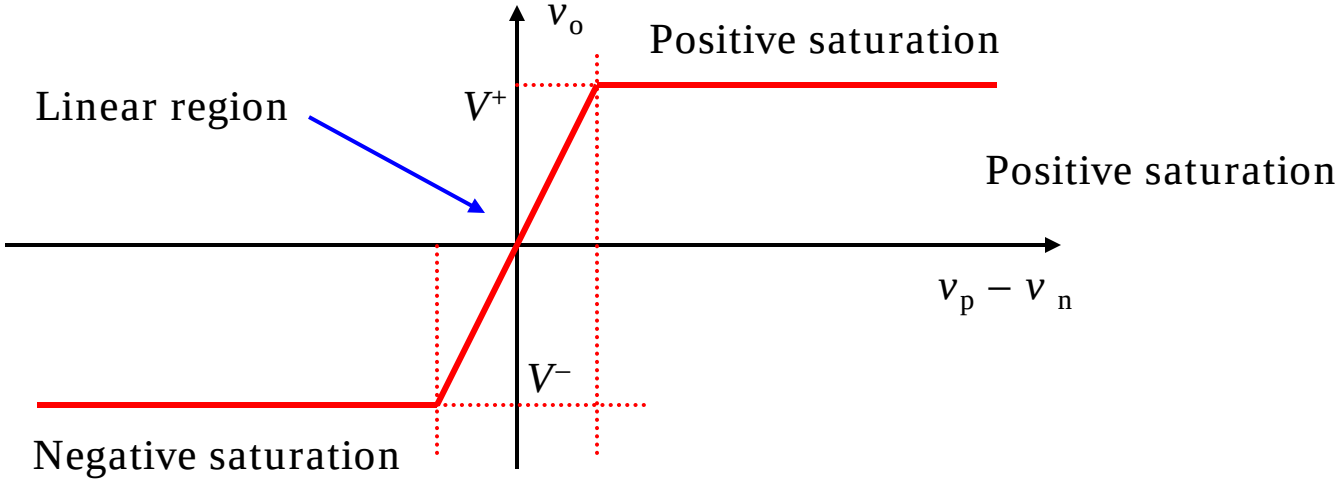
Feedbacks

Op Amp/Voltage Transfer Characteristics

Two major regions:

- Linear region: amplification
- Saturation: switch





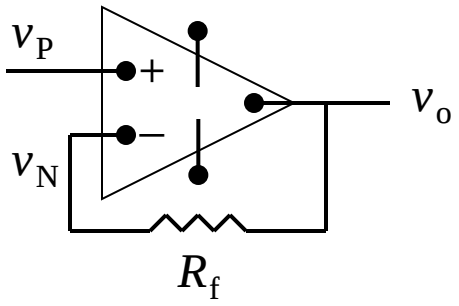
$$v_o = \begin{cases} V^-; \text{while } a(v_P - v_N) < V^- \text{ (negative saturation)} \\ a(v_P - v_N); \text{while } V^- \le a(v_P - v_N) \le V^+ \text{(linear)} \\ V^+; \text{while } a(v_P - v_N) > V^+ \text{ (positive saturation)} \end{cases}$$

Negative Feedback

Negative feedback can be used to ensure operation in the linear region.

- If ($v_P - v_N$) is too positive, then v_o increases, and thus v_N is increased by feedback loop.

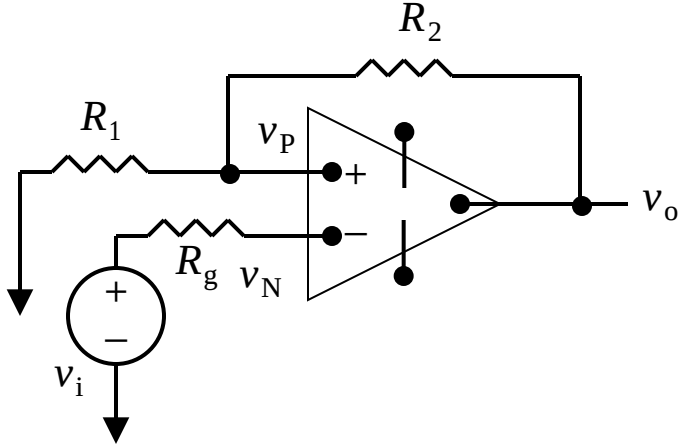
- If ($v_P - v_N$) is too negative, then v_o decreases, and thus v_N is reduced by feedback loop.



Positive Feedback

Use positive feedback op amp as shown:

$$v_P = v_O \frac{R_1}{R_1 + R_2}$$



$$\text{If } v_O = V_{OH}; \text{then } v_P = V_{OH} \frac{R_1}{R_1 + R_2} = V_{TH}$$

Valid as long as $v_i < V_{TH}$

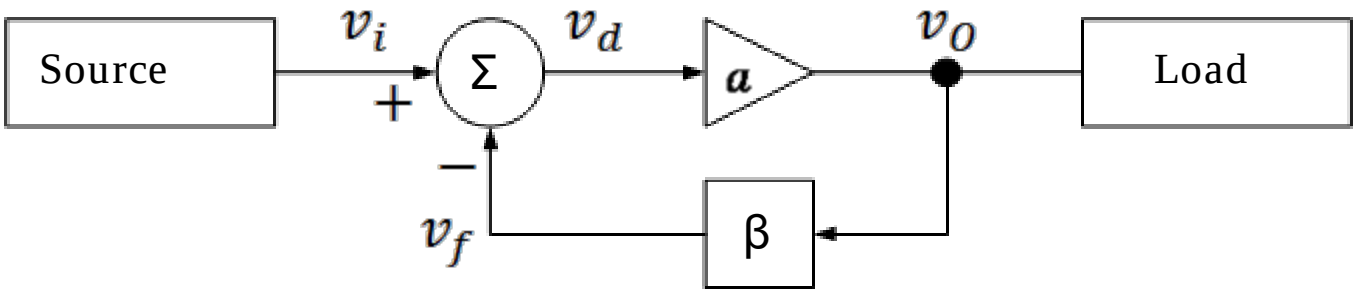
$$\text{If } v_O = V_{OL}; \text{then } v_P = V_{OL} \frac{R_1}{R_1 + R_2} = V_{TL}$$

Valid as long as $v_i > V_{TL}$

V_{OH}: voltage output high
V_{OL}: voltage output low
V_{TH}: threshold high voltage
V_{TL}: threshold low voltage

Negative Feedback

Negative Feedback



An amplifier accepts the signal v_d and yields the output signal

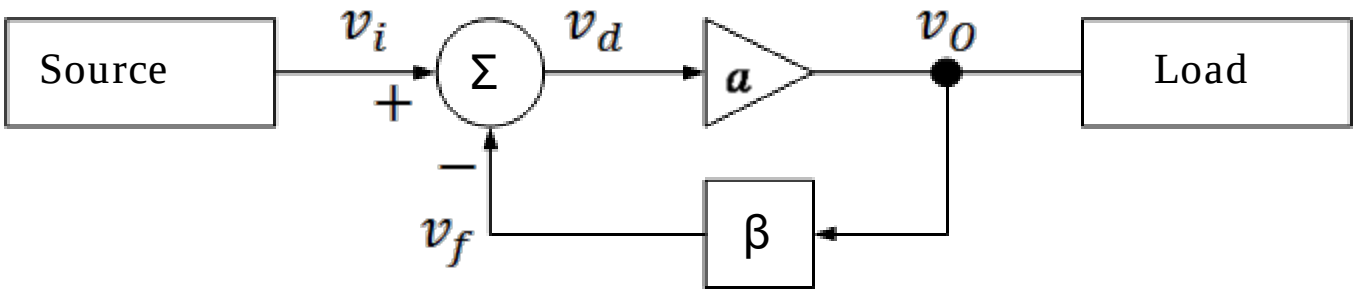
$$v_O = av_d$$

A feedback network, which samples v_o and produces the feedback signal

$$v_f = \beta v_o$$

where β, the gain of feedback network, is called the feedback factor of the circuit.

Negative Feedback



A summing network, denoted ∑, which generates the difference

$$v_d = v_i - v_f$$

Eliminate v_f and v_d from above equations, yields

$$A = \frac{v_O}{v_i} = \frac{a}{1 + a\beta}$$

where *A* is called closed-loop gain of the circuit.

Negative Feedback

As a signal propagates around the loop, the overall gain is

$$\text{overall gain } a \times \beta \times (-1) = -a\beta$$

The negative value is called loop gain, T

$$T = a\beta \text{ then } A = \frac{1}{\beta}\frac{T}{1+T}$$

Letting T$\rightarrow\infty$, yields the ideal situation,

$$A_{ideal} = \lim_{T\to\infty} A = \frac{1}{\beta}$$

A becomes independent of a.

Negative Feedback

We can express the closed-loop gain A as

$$A = A_{ideal} \times \frac{1}{1 + {}^{1}/_{T}} \qquad since\ A = \frac{1}{\beta}\frac{T}{1+T}$$

And reduced signal v_d as

$$v_d = \frac{v_O}{a} = \left(\frac{A}{a}\right) v_i = \frac{1}{1+T} v_i \quad \text{Since } A = \frac{a}{1+a\beta}$$

The feedback signal v_f as

$$v_f = \beta v_O = \beta A v_i = \frac{1}{1 + {}^{1}/_{T}} v_i \qquad since\ A = \frac{1}{\beta}\frac{T}{1+T}$$

As $T \to \infty$, $v_d = 0$ and $v_f = v_i$.

Virtual short

Negative Feedback: Sensitivity

Sensitivity

Investigate how variations in the open-loop gain a affects the closed-loop gain A,

$$\frac{\partial A}{\partial a} = \frac{1 + a\beta - a\beta}{(1 + a\beta)^2} = \frac{1}{1 + a\beta} \times \frac{1}{1 + a\beta} \qquad A = \frac{v_O}{v_i} = \frac{a}{1 + a\beta}$$

$$= \left(\frac{A}{a}\right) \frac{1}{1 + T}$$

As $T \to \infty$,

$$\Rightarrow \frac{\partial A}{A} = \frac{\partial a}{a} \frac{1}{1 + T} \to 0$$

Loop gain T desensitizes closed-loop gain A, to variations in the open-loop gain a, which are typically large.

Sensitivity/ Closed Loop Gain and Feedback Factor

From $A = \frac{1}{\beta}\frac{T}{1+T}$

$$\Rightarrow \frac{\partial A}{\partial \beta} = -\frac{1}{\beta^2}\frac{T}{1+T}$$ Also $A\beta = \frac{T}{1+T}$

This yields the relationship between closed-loop gain and feedback factor as shown below:

$$\frac{\partial A}{\partial \beta} \Rightarrow -\frac{A}{\beta} \Rightarrow \frac{\partial A}{A} = -\frac{\partial \beta}{\beta}$$

Change in β directly affects closed-loop gain.

Sensitivity/ Closed Loop Gain and Feedback Factor

Example: A negative feedback circuit has a = 10^5 and $\beta = 10^{-3}$.

Estimate the percentage change in A brought about by a $\pm$ 10% change in a.

$$\frac{\partial A}{A} = \frac{\partial a}{a}\frac{1}{1+T}$$

Sensitivity/ Closed Loop Gain and Feedback Factor

Example: A negative feedback circuit has a = 10^5 and β = 10^{-3}.

Estimate the percentage change in A brought about by a ± 10% change in a.

Solution:

From the relation $\frac{\Delta A}{A} \cong \frac{1}{1+T}\frac{\Delta a}{a}$

$\Rightarrow$ try to solve $\frac{\Delta A}{A}$ with $\frac{\Delta a}{a} = \pm 10\%.$

$$\frac{\Delta A}{A} = \pm 10\% \times \frac{1}{1+T} = \pm 10\% \times \frac{1}{1+a\beta} = \pm 10\% \times \frac{1}{1+100}$$

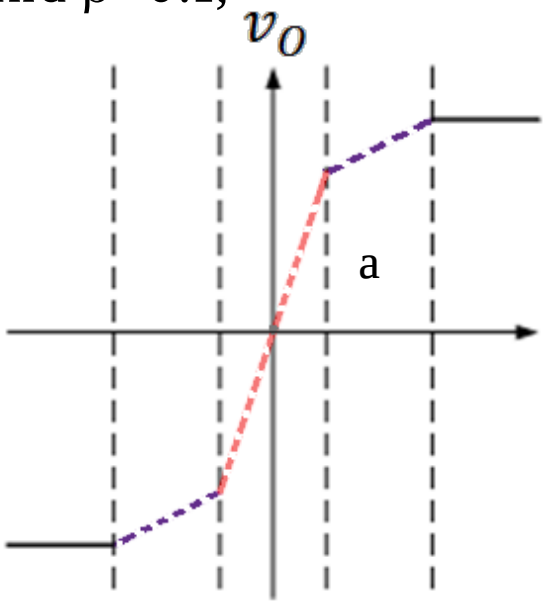
$\cong \pm 0.1\% \leftarrow$ desensitized.

Negative Feedback: Gain Linearity

Gain Linearity

Voltage transfer characteristic, v_O vs. v_d plot, is a convenient way to visualizing the transfer characteristic of an amplifier.

Example with an amplifier with nonlinear distortion gain regions, and β=0.1,

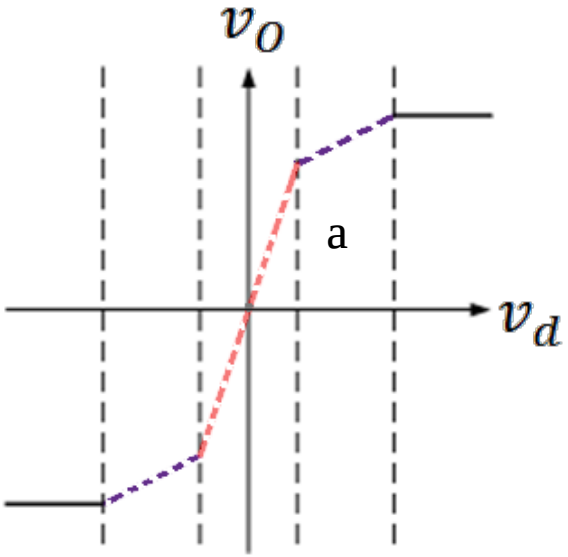


Purple (darker dotted) line: open-loop gain a = 100 V/V

Red (lighter dotted) line: open-loop gain a = 1000 V/V

Open-loop gain VTC
Variation of Vo vs Vd due to changes of a from 100V/V to 1000V/V

Gain Linearity



Open-loop gain VTC

Purple (darker dotted) line:
open-loop gain a = 100 V/V

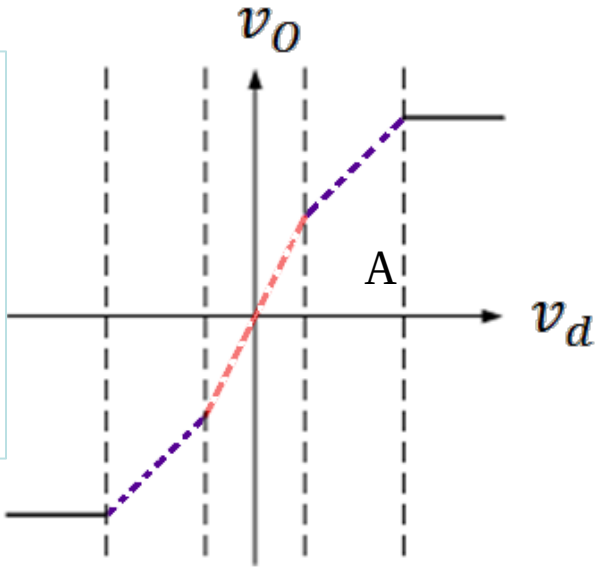
Red (lighter dotted) line:
open-loop gain a = 1000 V/V

From $A = \frac{a}{1 + a\beta}$

Purple (darker dotted) line: $A = \frac{100}{1 + 10} = 9.09 \frac{V}{V}$

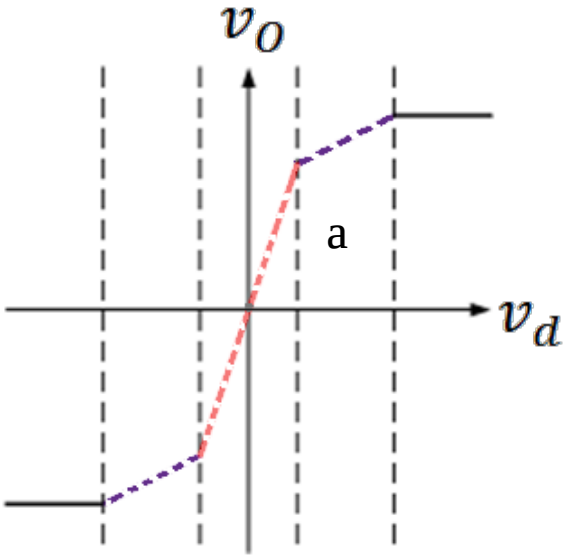
Red (lighter dotted) line: $A = \frac{1000}{1 + 100} = 9.90 \frac{V}{V}$

Much smaller close-loop gain VTC variation of Vo vs Vd due to changes of a from 100V/V to 1000V/V

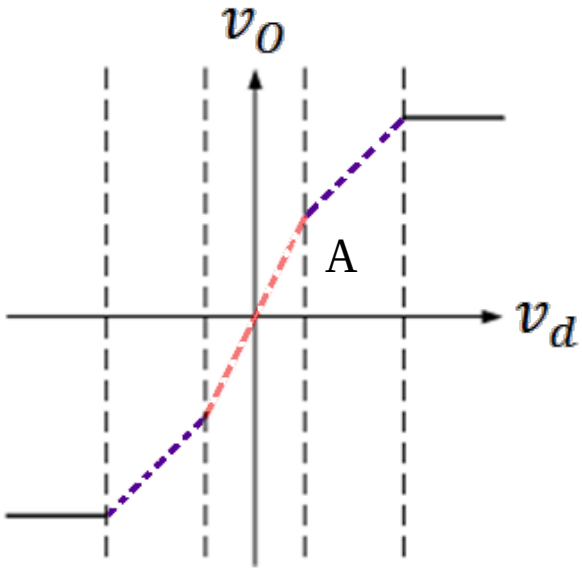


Closed-loop gain VTC

Gain Linearity



Open-loop gain VTC



Closed-loop gain VTC

Purple (darker dotted) line:
open-loop gain a = 100 V/V

Red (lighter dotted) line:
open-loop gain a = 1000 V/V

Purple (darker dotted) line:
closed-loop gain A = 9.09V/V

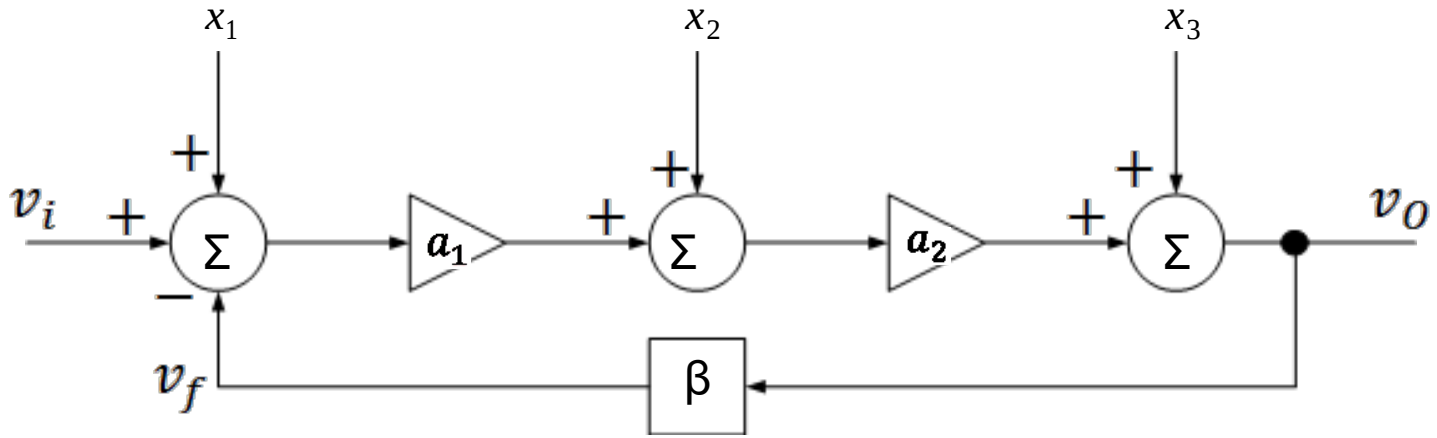
Red (lighter dotted) line:
closed-loop gain A = 9.90 V/V

10 X variation in a is around 10% variation in A,
which is much less distortion.

Negative Feedback: Noise Suppression

Noise Suppression

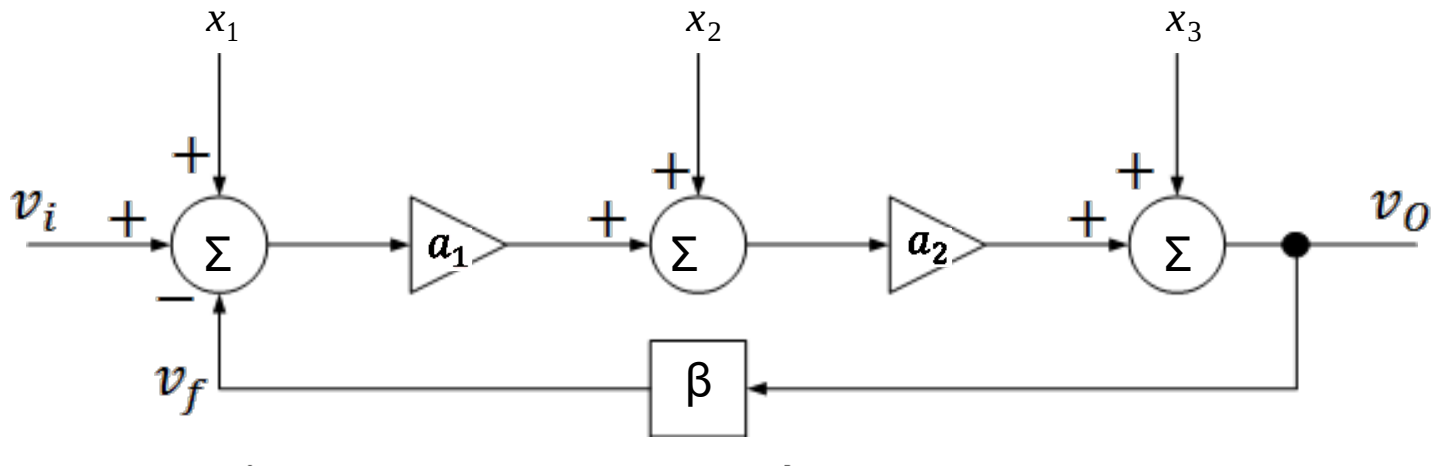
Negative feedback provides a means also for suppression circuit sensitivity to certain types of noise.



Noise sources include

- x_1: input offset errors, input noise
- x_2: power supply noise (hum)
- x_3: load changes, quantization noise, etc.

Noise Suppression



$$v_o = x_3 + a_2(x_2 + a_1(v_i + x_1 - \beta v_o))$$

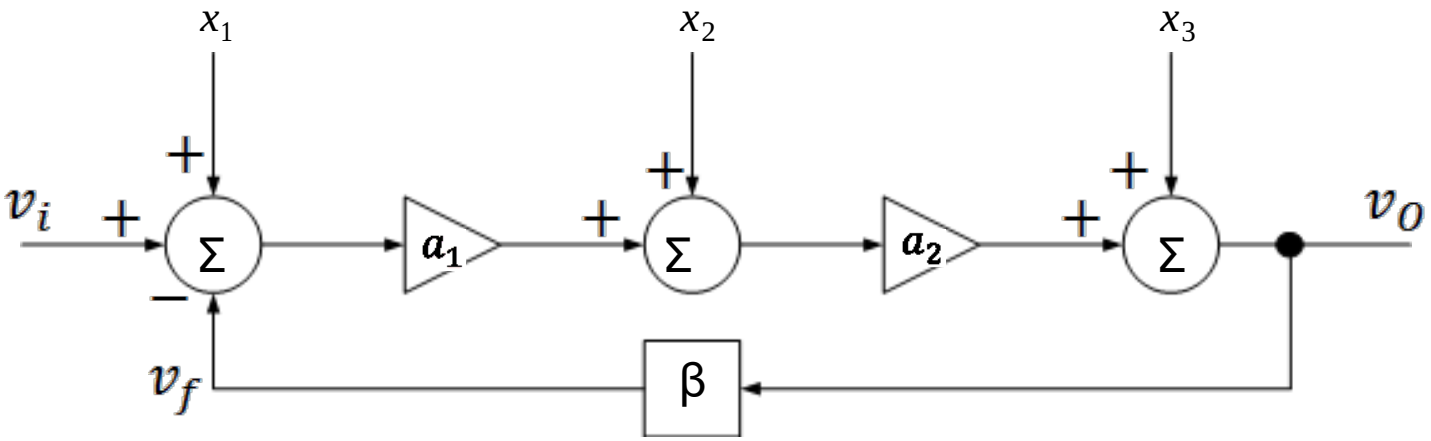
$$\Rightarrow \boldsymbol{v_O(1 + a_1 a_2 \beta) = a_1 a_2 (v_i + x_1) + a_2 x_2 + x_3}$$

Rearrange the equation above:

$$\boldsymbol{v_O = \frac{a_1 a_2}{1 + a_1 a_2 \beta}(v_i + x_1 + \frac{x_2}{a_1} + \frac{x_3}{a_1 a_2})}$$

This yields $\boldsymbol{v_O \approx \frac{1}{\beta}(v_i + x_1 + \frac{x_2}{a_1} + \frac{x_3}{a_1 a_2})}$

Noise Suppression

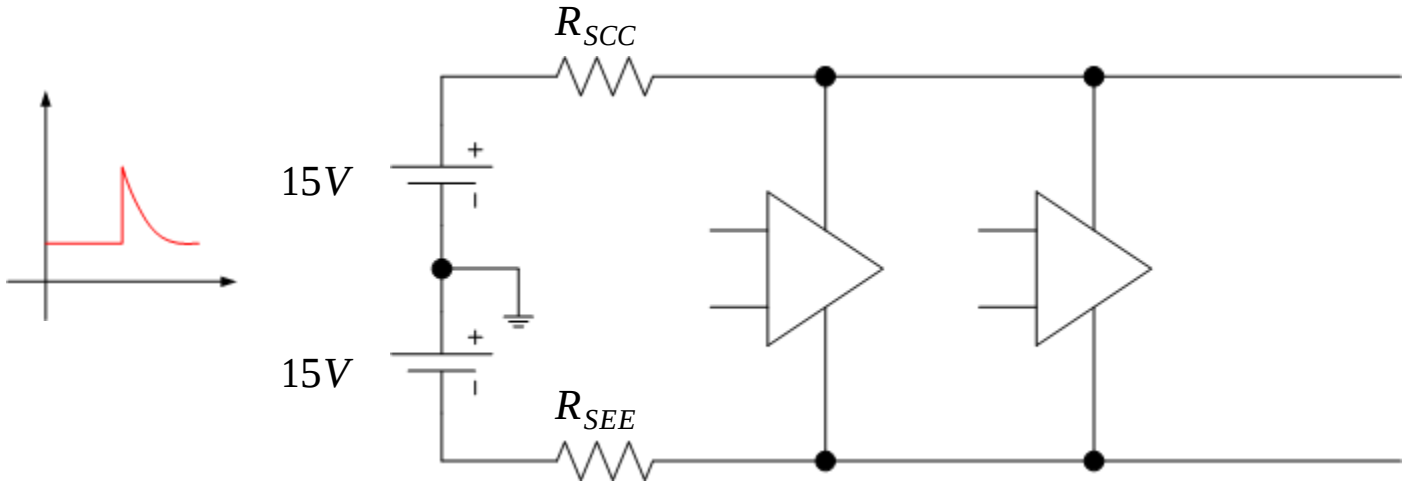


$$v_O \approx \frac{1}{\beta}\left(v_i + x_1 + \frac{x_2}{a_1} + \frac{x_3}{a_1 a_2}\right)$$

This shows the noise x_2 and x_3 have been reduced by the factors of a_1 and a_1a_2, respectively. Noise sources inside loop are suppressed.

Op Amp Powering

Op Amp Powering

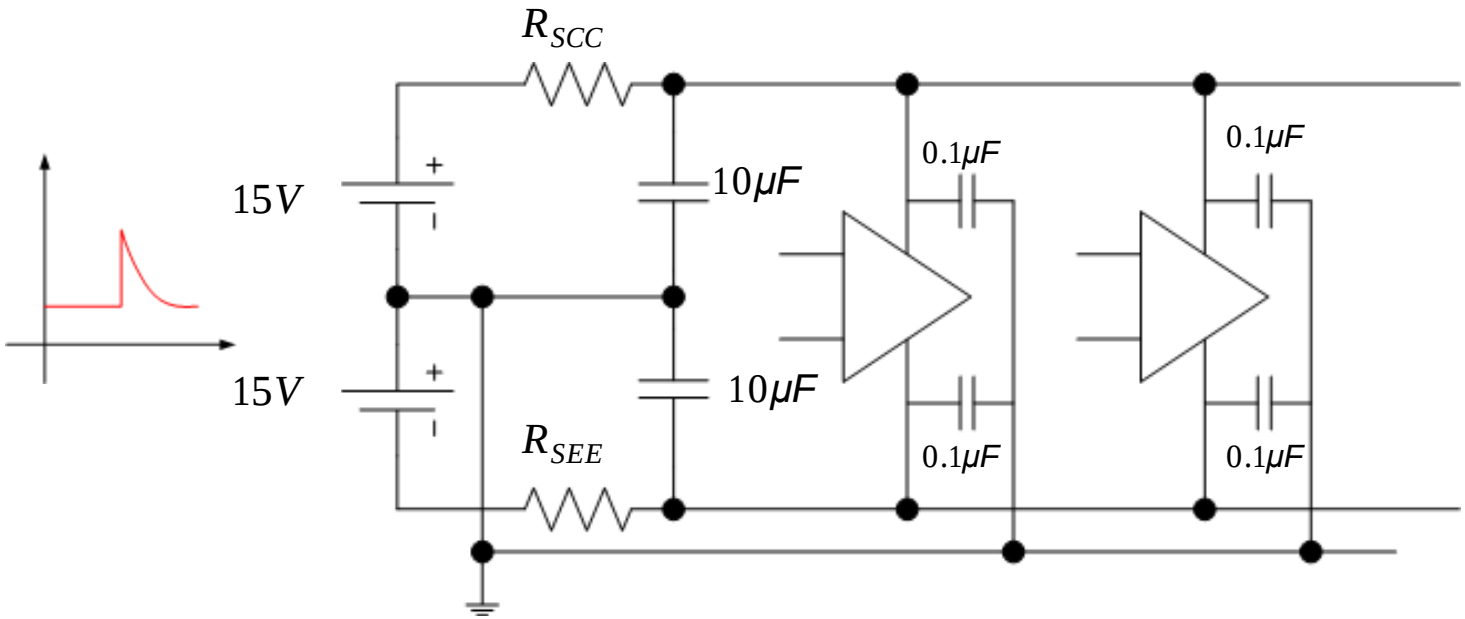


If a spike or noise generated by the sources, the disturbing signal will travel through the power rail, and may be amplified.

Op Amp Powering

Solution

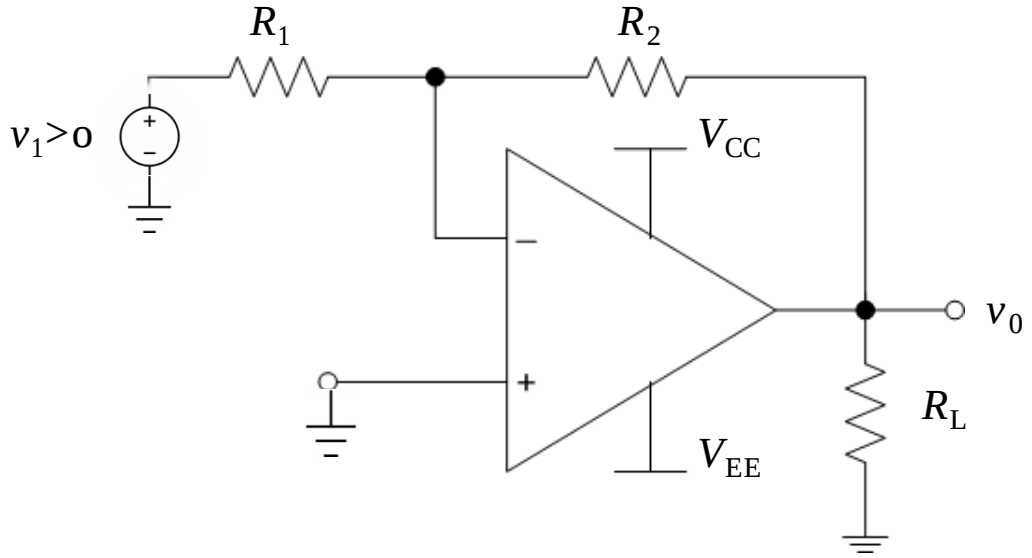
- Add power supply and connections to devices decoupling capacitors
- Around each op amp place 0.1 μF ceramic decoupling capacitor.
- At main power supply feed points place 10 μF tantalum decoupling capacitors.



Op Amp Powering: Current Flow and Power Dissipation

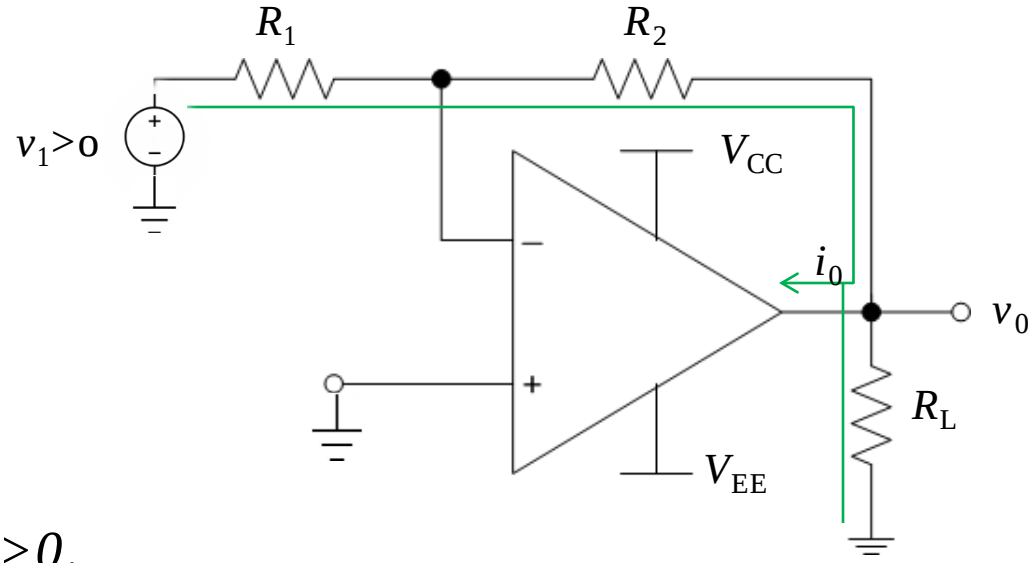
Current Flow and Power Dissipation

Use inverting amplifiers as an example,



- Four currents to consider,
  - i_{CC}: current flows from V_{CC} into op amp
  - i_{EE}: current flows from op amp into V_{EE}
  - i_O: current flows either in or out of op amp from output
  - I_Q: quiescent supply current from V_{CC} to V_{EE}

Current Flow and Power Dissipation



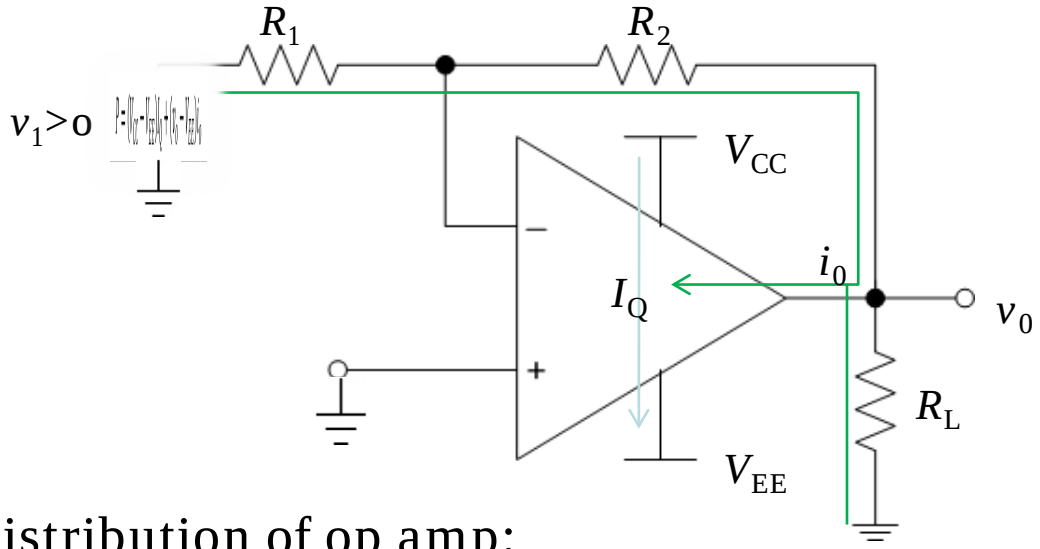
While $v_i > 0$,

$$v_o = -\frac{R_2}{R_1} v_1 < 0$$

i_O flows of output terminal of op amp and equals to

$$i_o = i_2 + i_L = i_1 + i_L = \frac{v_i}{R_1} + \frac{v_o}{R_L}$$

Current Flow and Power Dissipation



Power distribution of op amp:

$$P = (V_{CC} - V_{EE})I_Q + (v_o - V_{EE})i_o$$

I_Q is always existed, but smaller than i_O

Typically, select R_L<R_1 and R_2, then there is higher current through the load, but if selecting R_1 and R_2 too big, current might be too small to operate feedback.

Exercise

An inverting amplifier with R1=10 kohm, R2=20 kohm, and Vi=3V drives a 2 kohm load. Power supply is +/- 15V (a) Assuming I_Q=0.5mA, find i_{CC}, i_{EE} and i_o. (b) Find the power dissipated inside the op amp.

(a)

Vo=-(20/10)3=-6V

i_L=6/2=3mA

i2=i1=3/10=0.3mA

i_o=0.3+3=3.3mA

i_{CC}=I_Q=0.5mA

i_{EE}=i_{CC}+i_o=0.5+3.3=3.8mA

(b)

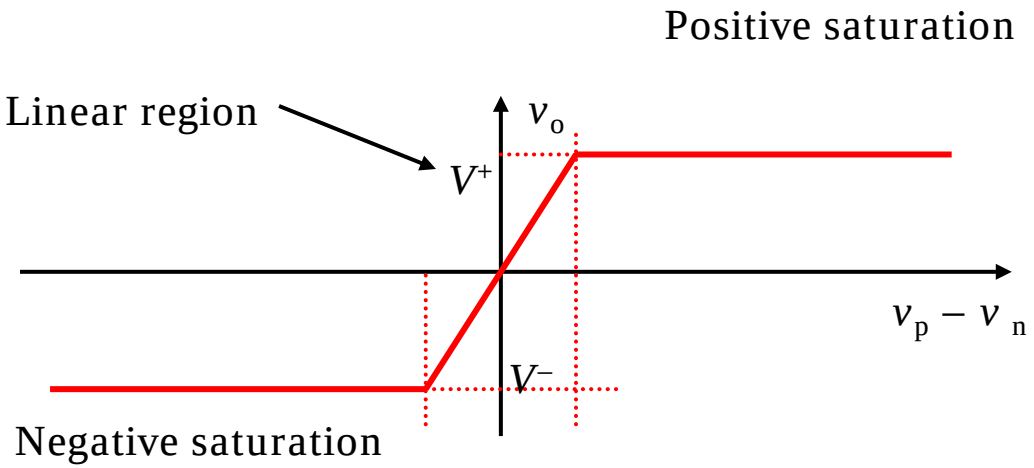
P=VI

P=(V_{CC}-V_{EE})I_Q+(V_o-V_{EE})i_o=30x0.5+[-6-(-15)]x3.3=44.7mW

Op Amp Powering: Output Saturation

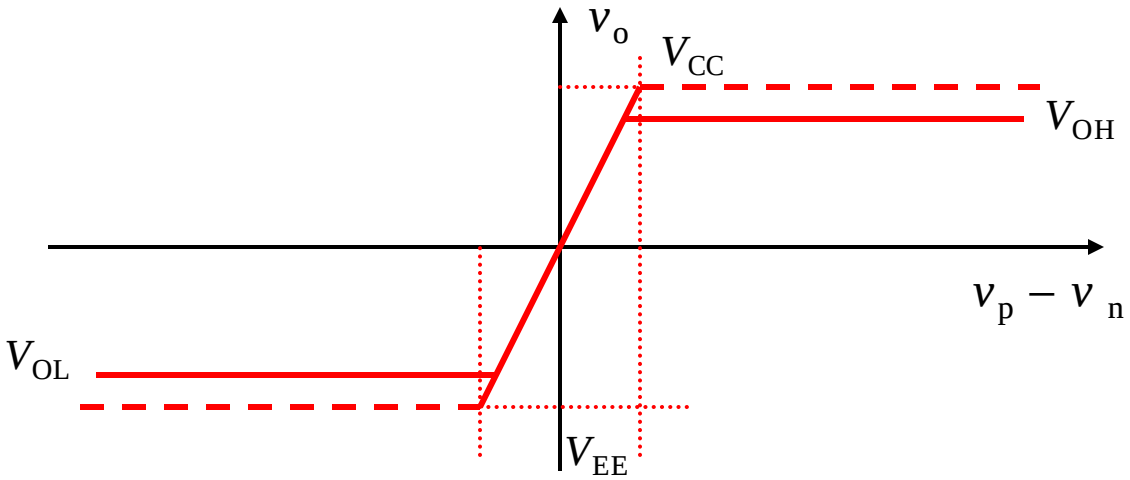
Output Saturation

Recall: It is best visualized in terms of the VTC which reveals linear and saturation regions.



$$v_o = \begin{cases} V^-; \text{while } a(v_p - v_N) < V^- \text{ (negative saturation)} \\ a(v_p - v_N); \text{while } V^- \leq a(v_p - v_N) \leq V^+ \text{(linear)} \\ V^+; \text{while } a(v_p - v_N) > V^+ \text{ (positive saturation)} \end{cases}$$

Output Saturation

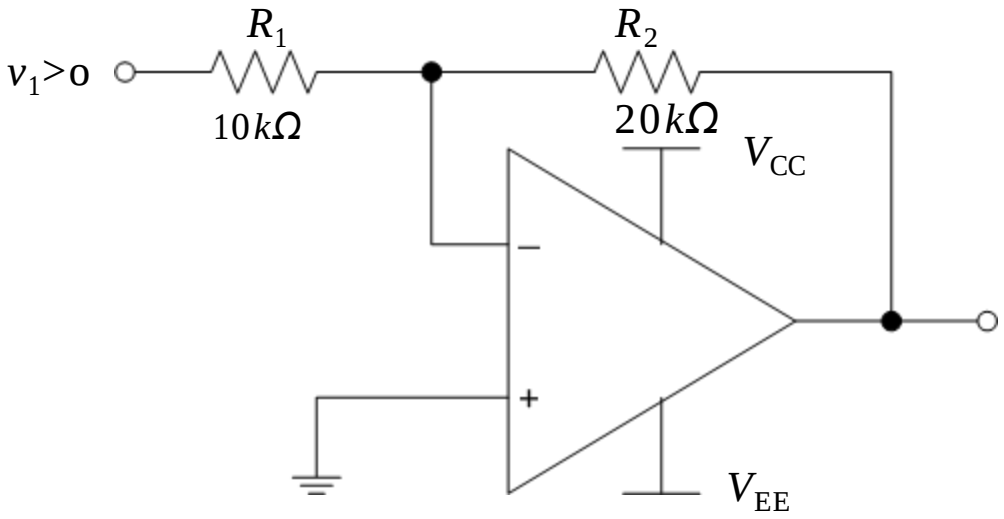


Voltage limit: V_{OH} and V_{OL} will not reach to the level V_{CC} and V_{EE}, bipolar op amp 741 for example, V_{OH} and V_{OL} are 2 V below V_{CC} and above V_{EE}.

Output Saturation

Voltage limit: V_{OH} and V_{OL} will not reach to the level V_{CC} and V_{EE}, bipolar op amp 741 for example, V_{OH} and V_{OL} are 2 V below V_{CC} and above V_{EE}.

Example: given the following op amp circuit, solve the output voltage:



$$V_{CC} = -V_{EE} = 15\ V,$$

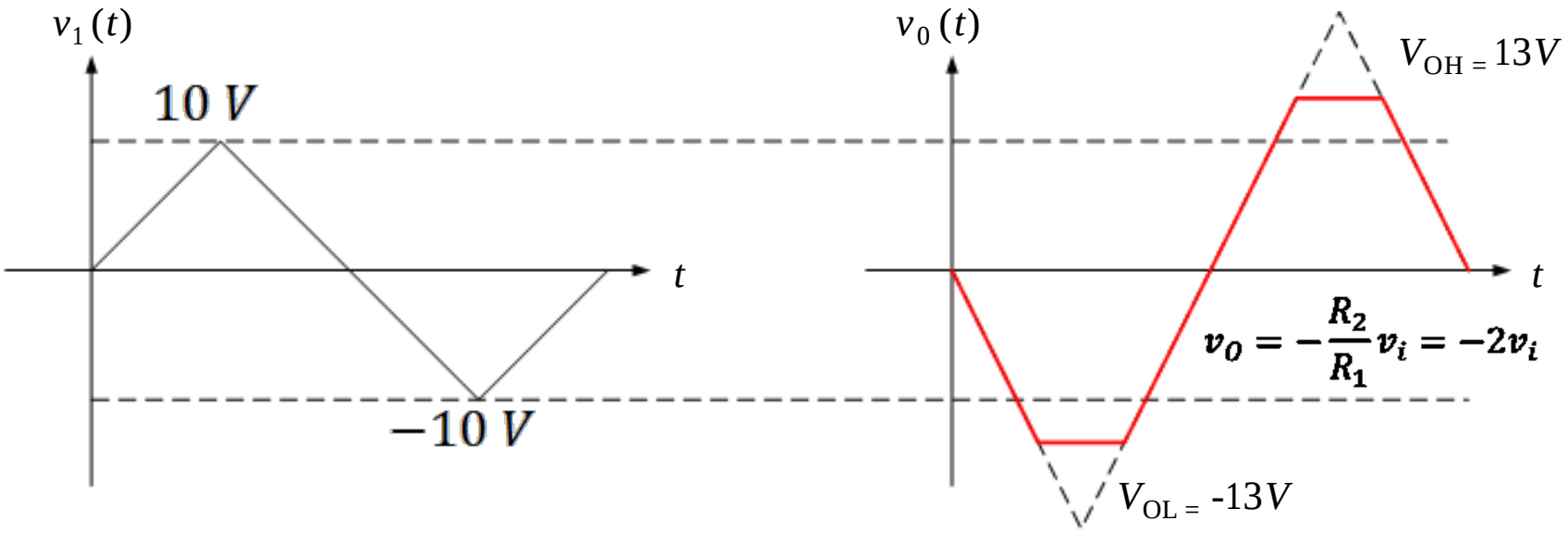
and $V_{OH} = -V_{OL} = 13\ V$

$$v_O = -\frac{R_2}{R_1} v_i = -2v_i$$

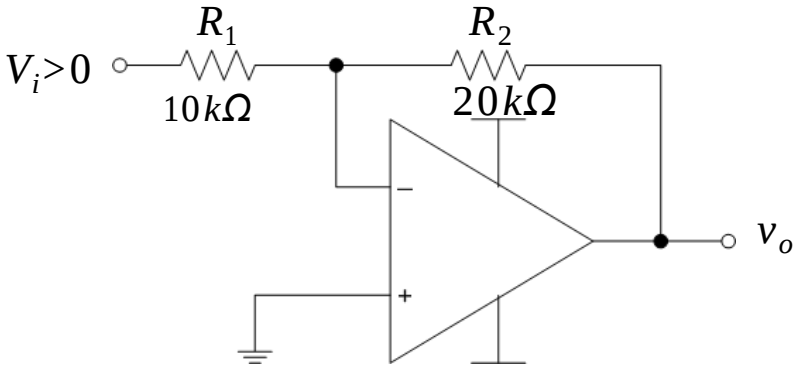
Output Saturation

Linear: Vi < +/- 6.5 V

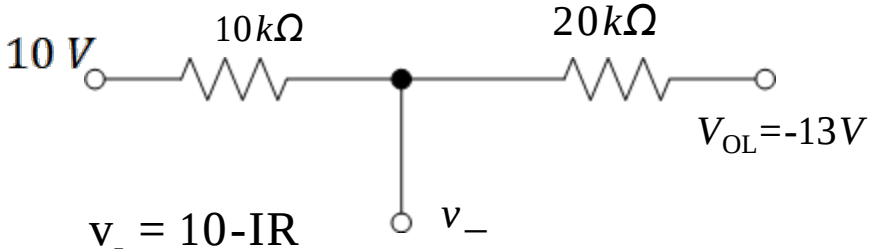
Solution:



Output Saturation



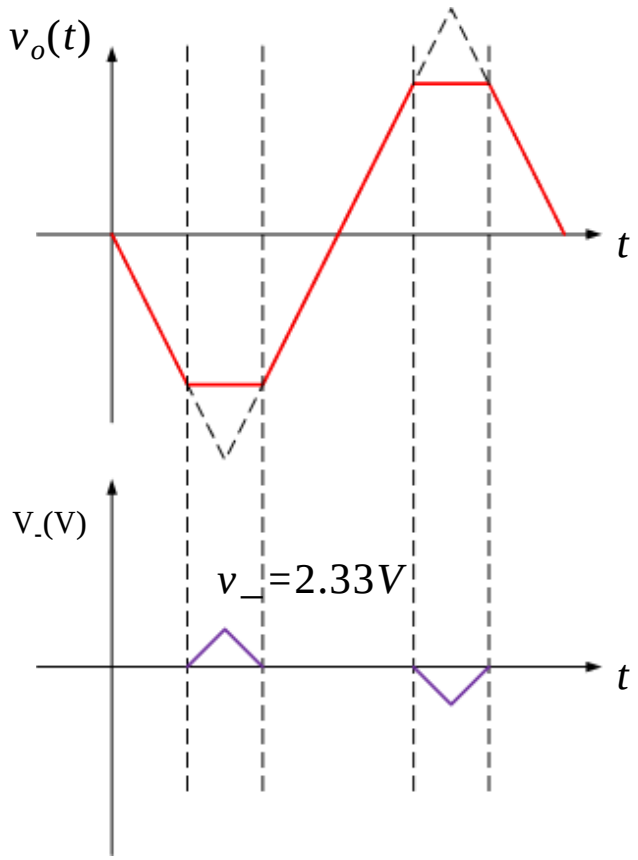
At positive peak : 10 V; v_O = -13V, consider the feedback loop:



v_- = 10-IR

$$v_- = 10 - \frac{10-(-13)}{10k+20k} \times 10k \Rightarrow v_- = 2.33V$$

V_- is no longer at virtual ground



This creates $v_- \neq v_+$ →No longer negative feedback control